# OPTICAL AND ELECTRON MICROSCOPY

## 1. INTRODUCTION AND HISTORICAL CONTEXT

### 1.1 The Two Fundamental Barriers in Biological Observation

The morphological and functional study of living matter has historically been conditioned by two physical limitations inherent to the biological scale:

- **The Size Barrier (Magnification):** Most eukaryotic cells have a diameter between 10 and 30 µm, while their organelles are situated on the submicrometer scale (0.1 – 1 µm) and macromolecular structures (protein complexes, nucleic acids, membranes) on the nanometer scale (2 – 25 nm). The human eye lacks the resolving power necessary to perceive these dimensions. To overcome this limitation, the use of microscopes is required. The conventional optical microscope achieves an image magnification of about 1,500 times, whereas the electron microscope reaches between 150,000 and 200,000 magnifications. The naked human eye has a resolution limit (resolving power) of approximately 0.1 mm (100 µm), allowing it to discern only structures such as the human oocyte (~120 µm), macroscopic histological sections, or large parasites such as helminths. Any structure smaller than this distance is perceived as blurry or as a single point if not mediated by lenses.
- **The Contrast Barrier (Transparency):** Cellular cytoplasm and the extracellular matrix space are composed of more than 70% water, combined with low optical density solutes. With few exceptions (such as hemoglobin or melanin), non-fixed and unstained cells absorb light homogeneously across the entire visible spectrum, behaving as pure phase objects and remaining virtually invisible and indistinguishable under an ordinary light microscope. To overcome this barrier, cell biology applies two main methodologies. The first is differential staining, which employs dyes with chromophore groups that modify the wavelength of light passing through the specimen, allowing structures to be seen in different colors. The second option is the use of special optical systems (such as the phase-contrast microscope), which detect and transform the small phase delays experienced by the wave when passing through or activating regions of different density and thickness into visible amplitude variations (brightness and light-dark contrasts) for the human eye.

### 1.2 Milestones in Instrumental and Conceptual Evolution

- **The Nimrud Lens (~750 BC):** A piece of polished rock crystal found in the Palace of Ashurnasirpal II (present-day Iraq) by Austen Henry Layard. Although there is controversy over whether it was a magnifying lens, a decorative element, or a ritual object for lighting fires by concentrating solar rays, it represents the oldest physical record of manipulating the curvature of a translucent optical medium.
- **The Origin of the Compound Microscope (16th–17th Centuries):** Zacharias Janssen and his father Hans Lippershey in the Netherlands (around 1590–1600) introduced the combined use of two compound lenses placed at opposite ends of a tube. Subsequently, Galileo Galilei (1609) and Johannes Kepler (1611) formalized the theoretical principles of the telescope and the compound microscope consisting of two convex lenses.
- **Robert Hooke and the Coining of the Term "Cell" (1665):** In his work Micrographia (1665), Hooke used a compound microscope equipped with a supplementary light source (a glass sphere filled with water that concentrated light from an oil lamp onto the specimen). Upon observing thin slices of plant cork (Quercus suber), he identified polyhedral pores that reminded him of monks' cells or "celdillas", naming them cellulae (cells). Hooke's compound microscope achieved approximately 50X magnification. The supplementary light source with the water sphere (condenser) was indispensable due to the faint flame of the oil lamp.

- **Anton van Leeuwenhoek and Simple Microscopy (1673–1680):** Unlike Hooke, Leeuwenhoek used single-lens or simple microscopes crafted with unprecedented artisanal polish. He achieved magnifications up to 250X and a resolution limit close to 1.4 µm. He was the first to describe free living cells, coining them "animalcules" (bacteria, protozoa, spermatozoa, and erythrocytes). The extreme polish Leeuwenhoek achieved on his small biconvex lenses allowed him to avoid the severe spherical and chromatic aberrations of the compound microscopes of his era, achieving a clarity that would not be surpassed until the 19th century.

# 2. CONVENTIONAL OPTICAL MICROSCOPY

## 2.1 Components of the Brightfield Microscope

The classic optical microscope forms its image based on the attenuation of visible light radiation after passing through stained tissue. Its architecture is divided into three functional subsystems:

### *A. Mechanical System*

**1. Base or Foot:** Provides stability to the center of gravity of the device.
**2. Arm or Pillar:** Vertical structure connecting the base to the optical tube and supporting the stage.
**3. Stage and Mechanical Slide Holder:** Flat surface perforated in the center. It holds the specimen using clips and moves it along X and Y axes via coaxial controls.
**4. Optical Tube:** Houses the eyepiece at the upper end and the revolving nosepiece holding the objectives at the lower end, maintaining a fixed tube focal distance (typically 160 mm or infinity-corrected).
**5. Coarse and Fine Focus Knobs:** Allow vertical focusing of the stage or optical tube. The coarse knob performs rapid movements for rough focusing; the fine knob makes subtle adjustments on the micrometer scale (precise focusing). The condenser also has a vertical movement control.

### *B. Illumination System*

**1. Light Source or Lamp:** Incandescent tungsten-halogen lamp or white LED diode with an electrical power regulator.
**2. Condenser:** System of converging lenses located under the stage whose function is to concentrate rays emitted by the light source to form an intense, homogeneous cone of light on the sample. Field diaphragms and filters are also included to optimize the beam.
**3. Aperture Diaphragm:** Iris device integrated into the condenser that regulates the diameter of the light cone, critically controlling contrast and depth of field.

### *C. Optical System*

**1. Objective:** The most critical optical component. It forms a real, inverted, and magnified image of the object. It is characterized by its nominal magnification (4X, 10X, 40X, 100X) and its Numerical Aperture (NA).
**2. Eyepiece (Ocular):** Lens responsible for collecting the intermediate image produced by the objective, acting as a magnifying glass to form a virtual, upright image relative to the intermediate image (inverted relative to the specimen), and magnified.

## 2.2 Resolution Limit, Resolving Power, and Abbe's Equation

### *A. Fundamental Definitions*

- **Resolving Power:** Qualitative ability of an optical system to reveal fine details, i.e., to display two closely spaced points or lines as distinct and independent structures.
- **Limit of Resolution (LR or d):** The quantitative minimum physical distance that must separate two points on a sample for the microscope to resolve them as two distinct entities. A smaller LR value equates to superior resolution quality.

### *B. Physical Basis: Diffraction and Airy Disk*

When light radiation passes through a biological structure or the objective lens aperture itself, optical diffraction occurs. The light rays bend and interfere with each other. Consequently, the image of a point light source is not projected as a perfect mathematical point, but as a bright central circular spot surrounded by attenuated concentric rings, known as the Airy Disk.

According to Rayleigh's Criterion, two neighboring points are considered resolved only when the center of the Airy disk of the first point coincides with the first intensity minimum of the Airy disk of the second point.

### *C. Ernst Abbe's Theoretical Equation (1873)*

Ernst Abbe mathematically formulated the resolution limit for incoherent illumination as follows:

$$LR\,(d) = \frac{k \times \lambda}{n \times \sin\alpha} = \frac{(0,61 \times \lambda)}{AN}$$

Where:

- **λ (Lambda):** Wavelength of light used (for the visible spectrum, the standard reference value is λ = 550 nm = 0.55 µm).
- **k:** Constant diffraction factor (typically 0.61 for incident illumination).
- **n:** Refractive index of the optical medium between the specimen and the front lens of the objective (n = 1.0 for air; n = 1.515 for immersion oil).
- **α (Alpha):** Half-angle of the maximum light cone collected by the front lens of the objective.
- **NA = n · sin α:** Numerical Aperture. Represents the lens's ability to collect the maximum amount of light diffracted by the sample.

### *D. Calculation of Maximum Resolution Limit*

Applying peak performance values for a visible light microscope (λ = 550 nm, NA = 1.40 with oil immersion):

$$LR\,(d) = \frac{(0,61 \times 0,55\mu m)}{1,40} \approx 0,24\mu m = 240nm$$

Thus, the insurmountable physical diffractive barrier of the classic brightfield microscope prevents distinguishing features at distances smaller than approximately 0.20 – 0.22 µm (200 – 220 nm). If light with a shorter λ were used, such as ultraviolet light, diffraction and interference would decrease, yielding a lower LR (better resolution). However, using (invisible) ultraviolet light requires special optical systems that convert the radiation into visible light after passing through the sample.

## 2.3 Comparison of Observation Scales and Capabilities

| Observation System | Resolution Limit (LR) | Visible Biological Structures |
|---|---|---|
| Naked Human Eye | 100 µm (0.1 mm) | Human oocyte (~120 µm), macroscopic histological sections, parasites (helminths). |
| Conventional Optical Microscope | 0.20 – 0.22 µm (200 nm) | Eukaryotic cells (10-100 µm), nuclei (~10 µm), nucleoli, larger mitochondria and lysosomes (~0.5-1 µm), bacteria (0.5-5 µm). |
| Scanning Electron Microscope (SEM) | 3 – 10 nm | Cell surface topography, erythrocytes, cilia, pollen, intracellular parasites. |
| Transmission Electron Microscope (TEM) | 0.1 - 0.2 nm (1 - 2 Å) in physical theory; 1 - 2 nm | Bilayer plasma membrane (7-10 nm), ribosomes (25 nm), viruses (30-100 |

| | | |
|---|---|---|
| | (10 - 20 Å) in biological tissue due to processing. | nm), cytoskeleton filaments (7-25 nm), macromolecules. |

## 2.4 Methods for Contrast Enhancement: Histological Stains

### *A. Physicochemical Foundation*

Biological specimens are processed via fixation (formalin) and paraffin embedding, then sectioned using a microtome (3–5 µm) and deparaffinized. Dye molecules feature two main components:

**1. Chromophore:** A chemical functional group possessing delocalized electrons capable of absorbing specific wavelengths in the visible light spectrum (λ = 400 – 700 nm), transmitting the complementary color.
**2. Auxochrome:** An ionically charged functional group (e.g., -COOH, -SO3-, -NH2+) that binds via ionic or covalent bonds to tissue molecular structures.

### *B. Classical Stain: Hematoxylin and Eosin (H&E)*

It is the routine diagnostic stain in anatomical pathology.

- **Hematoxylin (Basic Dye):** Behaves as a positively charged (cationic) molecule when complexed with potassium alum (Hematein). It exhibits high affinity for structures rich in negative charges (phosphate groups of nucleic acids). Structures stained by basic dyes are termed basophilic (e.g., nuclear chromatin DNA, ribosomal RNA). Staining color: dark blue-violet.
- **Eosin (Acidic Dye):** An anionic molecule (negatively charged). It binds ionically to positively charged amino groups (-NH3+) of cytoplasmic and extracellular proteins. Structures stained by acidic dyes are termed acidophilic or eosinophilic (e.g., cytoplasmic proteins, actin/myosin filaments, collagen fibers). Staining color: pink, red, or orange.

### *C. Specific Stains: Masson's Trichrome*

Used in diagnostic histology to quantitatively differentiate tissue fibrogenesis or fibrosis through the differential combination of three dyes:

- **Weigert's Hematoxylin:** Stains cell nuclei blue-black.
- **Biebrich Scarlet / Acid Fuchsin:** Stains cell cytoplasm, keratin, and muscle fibers intense red.
- **Aniline Blue / Light Green:** Has high affinity for dense collagen fibers of the extracellular matrix (stroma), staining them bright blue or green.

## 2.5 Contrast Without Staining: Phase-Contrast Microscopy

### *A. System Necessity*

Conventional histological stains are destructive (non-vital) dyes that cause cell death due to chemical processing (fixation, dehydration, xylene). To study dynamic cellular processes in vivo (mitosis, cell migration, amoeboid movement via pseudopodia), phase-contrast microscopy is required. It is routinely used in cell culture laboratories to observe living cells directly in their plastic flasks or dishes without damaging them.

### *B. Physical Mechanism of Interference*

Cellular structures (such as the nucleus, lipid droplets, or cytosol) exhibit subtle differences in refractive index (n) and thickness (e). When traversed by a wave train in phase, the light undergoes a phase shift of a quarter wavelength (δ = λ/4) relative to reference light rays passing through less dense regions. Because the human eye and digital sensors detect amplitude changes (brightness vs. darkness) rather than phase shifts, the microscope developed by Frits Zernike converts phase shifts into amplitude variations through the recombination of direct and diffracted light rays. Zernike was awarded the Nobel Prize in Physics in 1953 for inventing this microscope.

Essential components of the phase-contrast microscope:

1. **Annular Diaphragm (Condenser Ring):** Produces a hollow cone of direct illumination light.
2. **Phase Plate (in the Objective):** Contains an etched optical ring that further retards or accelerates the diffracted light by a quarter wavelength (λ/4) and reduces direct light intensity. Upon recombining at the rear focal plane, direct and diffracted light undergo destructive interference of half a wavelength (λ/4 + λ/4 = λ/2), canceling out or reinforcing in phase. This converts phase delays into visible light-dark contrast: denser cellular structures appear dark against a bright gray background.

# 3. FLUORESCENCE AND CONFOCAL MICROSCOPY

## 3.1 Fluorescence Microscopy

### *A. Physical Phenomenon of Fluorescence*

Fluorescence involves the absorption of radiant energy in photon form by a molecule (fluorophore) at a shorter wavelength (λexc), promoting its electrons from the ground state (S0) to excited singlet states (S1, S2). After losing a small portion of energy through non-radiative vibrational relaxation, the electron returns to the ground state by emitting a photon of lower energy and, consequently, a longer wavelength (λemi). Labeling may rely on autofluorescence (e.g., elastic fibers) or artificial conjugation (e.g., acridine orange staining nuclei or specific labeled antibodies).

### *B. Stokes Shift*

Defined as the spectral difference measured in nanometers between the peak of the absorption (excitation) spectrum and the peak of the emission spectrum of the fluorophore:

$$\Delta\lambda\, Stokes = \lambda emission - \lambda excitation$$

A large Stokes Shift is fundamental in microscopy to efficiently separate bright excitation light from the faint light emitted by the sample using optical filters without interference. Example of Fluorescein (FITC): Excites maximally with blue light at 494 nm and emits green fluorescent light at 521 nm. The broad shift allows optimal filtering without interference.

### *C. Components and Filters of the Epifluorescence Microscope*

In the epifluorescence configuration, the sample is illuminated from above through the objective lens itself. The optical system utilizes a Filter Cube containing:

1. **Excitation Filter:** Selects exclusively the specific spectral band required to excite the target fluorophore.
2. **Dichroic Mirror (Beam Splitter):** Glass coated with a dielectric film positioned at 45°. It reflects short excitation wavelengths toward the specimen while remaining completely transparent to longer emission wavelengths received from the tissue.
3. **Emission Filter (Barrier Filter):** Blocks any residual reflected excitation light and allows only photons emitted by the specimen to pass to the detector or camera.

## 3.2 Laser Scanning Confocal Microscopy (CLSM)

### *A. The Out-of-Focus Light Problem*

In conventional epifluorescence microscopy, the entire specimen is illuminated uniformly. Planes above and below the focal plane emit diffuse fluorescence light that reaches the detector, causing light contamination that blurs the image and severely degrades contrast and 3D resolution.

### *B. The Confocal Pinhole*

The confocal microscope (invented by Marvin Minsky) solves this problem by introducing a highly focused, intense point light source (a laser beam) and a small calibrated aperture or pinhole positioned in the conjugate focal emission plane. The laser passes through a first pinhole to focus onto a precise point in the specimen. Following excitation, emitted fluorescence passes through a second pinhole before reaching the detector (photomultiplier tube, PMT). Function of the Pinhole: Serves as a spatial filter physically blocking photons generated outside the focal plane of interest. Only rays originating from the exact focal point pass through the pinhole to reach the detector, dramatically enhancing image quality and sharpness.

### *C. Optical Sectioning and 3D Reconstruction*

By scanning the laser point along X and Y axes using galvanometric mirrors (scanning system) and systematically adjusting the stage height (Z-axis), the microscope acquires a series of extremely thin optical sections from a thick sample without destroying it (since lasers achieve deeper tissue penetration). Each section is termed a z-stack. Combining different stacks on a computer using image analysis software enables three-dimensional reconstruction. Using specialized software and deconvolution algorithms, interference signals from simultaneous multi-fluorophore labeling can be removed, allowing quantitative structural analysis.

# 4. ELECTRON MICROSCOPY

## 4.1 Physical Principles of Electron Optics

Due to the physical diffraction limit imposed by visible light wavelengths (~550 nm), the resolution limit of visible light microscopes cannot exceed 200 nm, preventing visualization of internal organelles. Electron microscopy bypasses this obstacle by replacing photons with a beam of accelerated electrons in a vacuum, dramatically increasing resolving power.

### *A. Louis de Broglie Wavelength*

According to wave-particle duality postulated by Louis de Broglie, a moving electron exhibits wave behavior with a wavelength (λe) dependent on its linear momentum. For an electron microscope operating at an accelerating voltage of 100 kV (100,000 V), the electron beam wavelength is merely 0.0037 nm (0.037 Å)—more than 100,000 times smaller than visible light.

### *B. Electromagnetic Lenses*

Since electrons cannot be refracted by glass, electron microscopes utilize electromagnetic coils (electromagnets). Passing a regulated electric current through the coils generates an axial magnetic field that deflects and focuses the trajectory of the electron beam.

## 4.2 Transmission Electron Microscope (TEM)

### *A. Architecture and Operation*

Consists of a cathode ray tube with a column approximately 1 meter long. At the top, a cathode (tungsten or wolfram filament) emits an electron beam driven by a high potential difference toward a perforated anode. The beam passes through the anode and is condensed by an electromagnetic coil onto the specimen (supported on a lentil-sized copper or nickel grid). The beam passes through the specimen and is focused by objective and projector coils onto a basal fluorescent screen. The fluorescent screen absorbs invisible electron radiation and emits visible light (radiation of longer λ) to form the image.

### *B. Vacuum and Sample Thickness*

Electron transmission requires a high vacuum (maintained by a vacuum pump system), meaning in vivo studies cannot be conducted with this technique. For the electron beam to pass through the specimen,

it must be extraordinarily thin: between 10 and 100 nm (or 0.05 µm). These sections are prepared by embedding tissue in hard plastic resins (polymerized by heat or UV) and sectioning with an ultramicrotome equipped with glass or diamond knives.

### *C. Contrast by Electron Scattering*

Contrast is achieved through differential absorption of heavy metal salts (an oxidizing fixative like osmium tetroxide, plus lead citrate and uranyl acetate stains) that bind to tissue structures based on chemical affinity.

- **Electron-dense (Dark) Regions:** High molecular weight structures retaining heavy metal salts that scatter or deflect electrons, preventing direct passage to the screen.
- **Electron-lucent (Light) Regions:** Areas absorbing few metal salts, allowing direct passage of the electron beam to the screen.

**Clinical Application:** Visualization of organelle ultrastructure, such as evaluating cilia in primary ciliary dyskinesia (which causes infertility, pregnancy complications, deafness, and recurrent respiratory infections like bronchitis) for ultrastructural diagnostic assessment.

## 4.3 Scanning Electron Microscope (SEM)

### *A. Scanning Principle and Metallization*

Used to observe cell and tissue surfaces at high magnification and resolution. Solid specimens of considerable size (such as whole tissue fragments or intact organisms) retain their volume and undergo a metallization process, coating them with a nanometer-thin film of conductive metal (gold or platinum). The primary electron beam scans the surface across three spatial dimensions.

### *B. Secondary Electrons and Detector*

The impact of the primary beam on the metalized specimen ejects peripheral orbital electrons from sample atoms (termed secondary electrons). The quantity emitted depends on the primary beam's angle of incidence. These secondary electrons are collected by an angular detector (sensor) and projected onto a monitor, generating a three-dimensional topographic image with shadows and metallic appearance. The beam does not penetrate the sample, but scans it. Typical magnifications: 30,000X.

## 4.4 Comparative Table: TEM vs. SEM

| Parameter / Feature | Transmission Electron Microscope (TEM) | Scanning Electron Microscope (SEM) |
|---|---|---|
| Image Nature | Flat, two-dimensional (2D) view of internal ultrastructure. | Three-dimensional (3D), topographic surface view. |
| Electron Trajectory | Beam passes through the specimen (transmission). | Beam scans the surface of the specimen (scanning). |
| Sample Thickness | Ultra-thin sections (50 – 90 nm) in epoxy resin. | Larger bulk specimens coated in gold/platinum. |
| Typical Resolution | 0.1 – 0.5 nm (1 – 5 Å); 10-20 Å in biological tissue due to preparation. | 30 – 100 Å |

# 5. ADVANCED BIOMEDICAL TECHNIQUES

## 5.1 Immunodetection of proteins

### *A. Physicochemical Principle and Immunoglobulins*

Technique based on the high specificity and affinity of binding between an immunoglobulin (antibody) and its target protein epitope (antigen) in tissue. Immunoglobulins are proteins composed of four polypeptide chains (two heavy and two light chains). The variable Fab region (antigen-binding) contains the antigen binding site (recognizing a specific epitope, which is a subregion of the full protein). The constant Fc region (crystallizable) is shared among all antibodies of the same class and species (e.g., mouse Fc differs from human Fc, enabling secondary antibody selectivity).

### *B. Direct and Indirect Detection Methods*

**Direct Method:** The primary antibody (recognizing the antigen of interest) is directly conjugated via its Fc region to a detectable molecule (enzyme, fluorophore, gold particle). It is the simplest single-step method but offers limited signal amplification.

**Indirect Method:** The primary antibody is unlabeled. A conjugated secondary antibody directed against the Fc region of the primary antibody is used (its Fab recognizes the primary antibody's Fc). This provides massive signal amplification (multiple secondary antibodies bind to one primary antibody) and is the most widely used approach in clinical diagnostics due to its technical convenience and versatility. Additionally, it provides greater signal amplification than direct methods. Like primary antibodies in direct detection, secondary antibodies can be conjugated to various detectable molecules (enzymes, fluorophores, gold particles).

### *C. Immunohistochemistry (IHC) Protocol*

In anatomical pathology diagnostic laboratories, indirect enzymatic labeling using the peroxidase enzyme represents the standard protocol because it requires no complex or expensive equipment for execution and visualization.

**1.** Tissue deparaffinization and rehydration.
**2.** Antigen retrieval (using heat and controlled pH) to unmask epitopes cross-linked by formaldehyde.
**3.** Endogenous peroxidase blocking to prevent false positives.
**4.** Incubation with the primary antibody specific to the target protein.
**5.** PBS buffer washes to remove unbound antibodies.
**6.** Incubation with the secondary antibody conjugated to horseradish peroxidase (HRP).
**7.** PBS buffer washes.
**8.** Enzymatic chromogenic revelation: hydrogen peroxide ($H_2O_2$) substrate is added in the presence of 3,3'-diaminobenzidine (DAB). Peroxidase reduces $H_2O_2$ and oxidizes DAB, forming a brown insoluble precipitate at the binding site. If the antigen is present, cells stain brown.
**9.** Counterstaining with Hematoxylin to stain nuclei blue-violet, followed by dehydration, mounting, and observation under a light microscope.